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Why Does Mobile Signal Strength Vary? An Analysis of the Shielding Effect from Maxwell's Equations.

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1 Abstract

1.1 Purpose:

Signal intensity often fluctuates in everyday environments. This study aims to investigate the key factors influencing signal strength within enclosed spaces. Specifically, we seek to answer: What are the primary factors that affect the transmission rate of electromagnetic plane waves—or, equivalently, what determines the effectiveness of electromagnetic shielding?

1.2 Methods:

We adopt a multi-faceted approach involving theoretical analysis for conductive materials, simulation experiments for both conductive and non-conductive scenarios, and real-world measurements to validate our hypotheses.

1.3 Key Findings:

Our results show that the electromagnetic properties—specifically the electric conductivity and magnetic permeability—of materials encountered by the waves are the most critical factors in determining shielding effectiveness. However, contrary to our initial assumptions, we found that factors beyond material composition, such as physical structure (e.g., gaps in elevator doors), multipath propagation, and environmental complexity, also significantly affect signal reception.

1.4 Conclusion:

In typical scenarios, shielding effectiveness is the dominant factor affecting signal strength in enclosed environments. Nevertheless, due to the inherent complexity of spatial geometry, wave behavior, and communication-related phenomena, accurately assessing signal intensity requires more precise measurement tools and refined theoretical models.

2 Introduction

In practice, people notice that mobile phone signal quality fluctuates—it is sometimes strong and other times weak. What particularly caught our attention was why thin, gapped (light-permeable, air-permeable) elevator doors can significantly weaken the signal, while thicker, wind-proof and rain-proof building structures have almost no effect on signal strength. In this study, we examine two common scenarios to explore the main physical reason from electromagnetic theories: an enclosed elevator and a standard room within a building such as a library discussion room or a college classroom.

Firstly, we utilize the Maxwell's equations to get the shielding effectiveness (SE) of different materials. The SE is a critical concept in the field of electromagnetic compatibility which can be expressed as a sum of three terms, $SE = A + R + B$, all terms in decibels unit. The more the value of SE is, the stronger the signal attenuation. In this report, we simply treat the signal as a transverse electromagnetic plane wave (TEM), which means the electric field is perpendicular to the magnetic field everywhere and the propagation direction of wave is perpendicular to the plane formed by B and E. **Expanding on the theoretical calculations**, we make use of a simulation tool to verify the calculation correctness and get some graphical representations that can clearly show the SE in the two scenarios. **Moreover**, we measure the real data in the two scenarios through RSRP values that can be converted to the SE values after simply subtracting two values, one from the space with no shielding and the other one with shielding. In this part, since we notice the fluctuations in the same scene, we take the multi-path effect into account to make the experimental data more reasonable. **Finally**, we offer some intuitive recommendations from the theories we have introduced for manufacturers of enclosed spaces to enhance signal transmission quality and provide advice for consumers seeking to improve signal quality in confined areas.

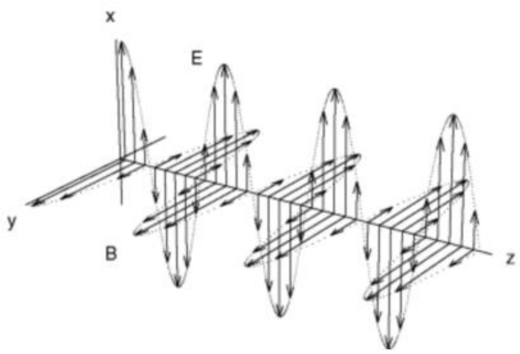

Fig. 1: TEM [10]
In the figure, along the z axis, there are many vectors representing the $\overrightarrow{E}$ or $\overrightarrow{B}$. In detail, this wave is propagating along the positive direction of z axis, because of $\overrightarrow{E} \times \overrightarrow{B}$.

3 SE Derivation from Maxwell's Equation

In this part, we start from considering the situation that of what will happen after the TEM encounters a medium interface. Then, we can define the SE from electric field E view.

3.1 What happens when TEM encounters an interface?

3.1.1 Intuitive View

Just like light encountering an interface will produce reflection and refraction, when a TEM encounters a medium interface, as illustrated as Figure 1, a part of the wave will reflect back to medium 1, and the other part will go through the medium 2.

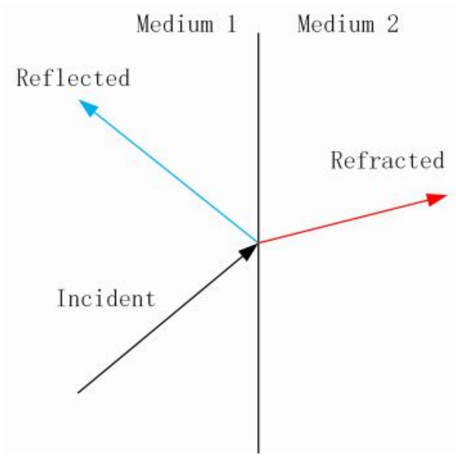

Fig. 2: simple

Unlike the light example, we have to consider the second "reflection and refraction" problem because of the definite thickness of medium 2.

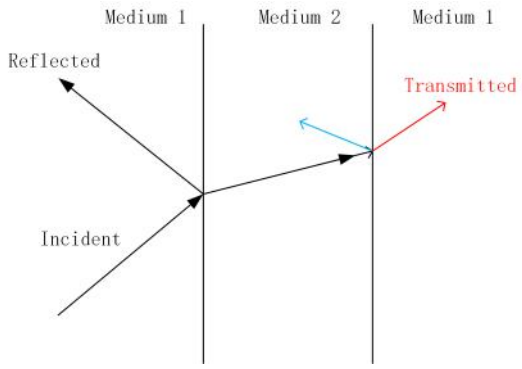

Fig. 3: thickness consideration

3.1.2 Qualitative analysis: What will eventually spread

From Fig. 2, we can observe that the wave is weakened twice at the two interfaces; this phenomenon is called reflection loss. Additionally, when the wave travels through medium 2 (which is assumed to be a substance other than air), it also experiences weakening due to absorption loss. There is a possibility that the second reflected wave will return to the second interface, allowing more wave energy to enter back into medium 1, as illustrated in Fig. 3. Beyond this second reflected wave returning to interface 2, additional reflected waves may also occur. The issue of what is known as multi-reflection will be discussed in Section "B Term" of 3.2.

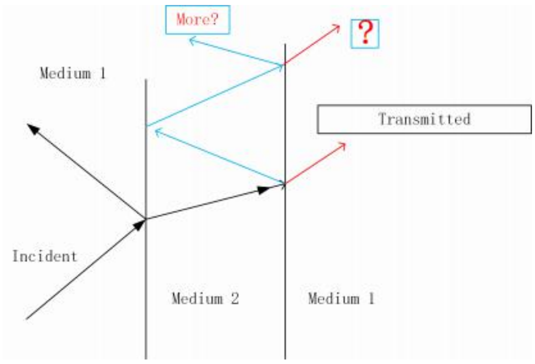

Fig. 4: More?

3.2 $SE = A + R + B$ Based on E Field

The following content (within section 3.2) is mainly modified from [3], [2], [6] and [5]. Our writing avoids plagiarism while ensuring accuracy based on these publications.

3.2.1 Wave Equation in the Media

Let $\overrightarrow{H} = \overrightarrow{B}/\mu$, considering the differential form of Maxwell's Ampere's law, with $\overrightarrow{J} = 0$, we have

$$\nabla \times \overrightarrow{H} = \epsilon \frac{\partial \overrightarrow{E}}{\partial t} \tag{1}$$

Suppose the wave travels in x direction,

$$\frac{\partial H_z}{\partial x} = -\epsilon \frac{\partial E_y}{\partial t} \tag{2}$$

Considering differential form of Faraday's law of electromagnetic induction,

$$\nabla \times \overrightarrow{E} = -\frac{\partial \overrightarrow{B}}{\partial t} \tag{3}$$

Suppose the wave travels in x direction,

$$\frac{\partial E_y}{\partial x} = -\mu \frac{\partial H_z}{\partial t} \tag{4}$$

Differentiate (2) respective to time, and differentiate (4) respective to distance, we have

$$\begin{cases} \frac{\partial^2 H_z}{\partial t \partial x} = -\epsilon \frac{\partial^2 E_y}{\partial t^2} \\ \frac{\partial^2 E_y}{\partial x^2} = -\mu \frac{\partial^2 H_z}{\partial x \partial t} \end{cases} \tag{5}$$

Combining these two equations, we have the wave equation in air

$$\frac{\partial^2 E_y}{\partial t^2} = \frac{1}{\mu\epsilon} \frac{\partial^2 E_y}{\partial x^2} \tag{6}$$

And the solution form is

$$E_y = E_0 sin(wt - \beta x) \tag{7}$$

In a lossy medium, $\sigma \neq 0$, from (2), we have

$$-\frac{\partial H_z}{\partial x} = \sigma E_y + \epsilon \frac{\partial E_y}{\partial t} \tag{8}$$

Converting (7) to $E_y = E_0 e^{jwt}$, and by (4), we have

$$\frac{\partial^2 E_y}{\partial x^2} = (jw\mu\sigma - w^2\mu\epsilon)E_y \tag{9}$$

If we denote $\gamma^2 = jw\mu\sigma - w^2\mu\epsilon$, i.e. $\frac{\partial^2 E_y}{\partial x^2} = \gamma^2 E_y$, the solution form is

$$E_y = E_0 e^{-\gamma x} \tag{10}$$

For conductor, $\gamma^2 = jw\mu\sigma$, after conversion, $\gamma = (1+j)\sqrt{\frac{w\mu\sigma}{2}}$. Finally,

$$E_y = e^{-x/\delta} e^{-j(x/\delta)}, \delta = \sqrt{2/w\mu\sigma} \tag{11}$$

If we omit the complex part of E_y, we have

$$E_y = E_0 e^{-x/\delta} \tag{12}$$

Through the (12), we can easily know the electric field intensity at any position x in the medium as shown in the Figure 5. **<u>Notice:</u>** There is a confusion about the E_0 in (12), in fact, applying it to the Fig. 5, we have to calculate the electric field after attenuation.

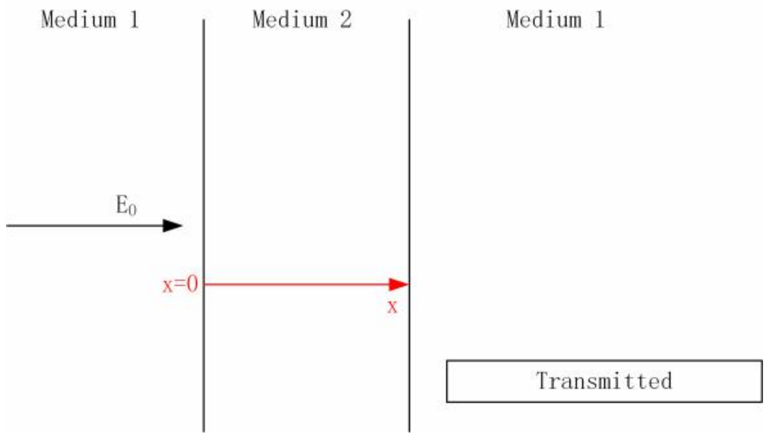

Fig. 5: Scenario

3.2.2 A and R Terms

In SE theory, the reduce effect of the electromagnetic field is in the form of absorption (A), reflection (R) and multi-reflection correction (B). In this section 3.2.2, we will illuminate the deductive process of A and R. In this section, we simply omit the B term consideration, and we will discuss it in section 3.2.3.

From publications, we have known that the finally transmitted proportion of electric field is dependent on the relationship between medium 1 and 2 (Figure 5) of the characteristic impedance. Specifically, the transmitted proportion is based on the continuity for both E field and B field. And the conclusion is, referring to the Fig. 5, if we define impedance of medium 2 as Z_2, of 1 as Z_1, and denote the electric field before the interface as E_{before}, after the interface as E_{after}, considering the first interface transmission:

$$k = \frac{E_{after}}{E_{before}} = \frac{2Z_2}{Z_1 + Z_2} \tag{13}$$

Referring to the Fig. 5, we define the E_1 as the quantity in the medium 2, and E_2 in the right-hand side medium 1. Suppose the thickness of the medium 2 is a,

$$E_1(x = 0) = E_0 k_{12} \tag{14}$$

$k_{12} = k$ of (13). At the $x = a$, applying (12),

$$E_1(x = a) = E_1(x = 0) \times e^{-a/\delta} \tag{15}$$

And the second interface, from medium 2 to medium 1,

$$E_2(a) = E_1(x = a) \times k_2 = E_1(x = a) \times \frac{2Z_1}{Z_1 + Z_2} \tag{16}$$

Combining the above three equations,

$$E_2(a) = E_0 \frac{4Z_1 Z_2}{(Z_1 + Z_2)^2} e^{-\frac{a}{\delta}} \tag{17}$$

Since the medium impedance has the formula of

$$Z = \sqrt{\frac{jw\mu}{\sigma + jw\epsilon}} \tag{18}$$

where μ inductance per unit length, ϵ capacitance per unit length and σ for conductance Here, considering good conductor, which means $\sigma >> w\epsilon$, the (18) reduces to

$$Z = \sqrt{\frac{jw\mu}{\sigma}} \angle 45^{\circ} \tag{19}$$

With good conductor medium 2 assumption one more time, we can reduce the (17) to

$$E_2(a) = E_0 \frac{4Z_2}{Z_1} e^{-\frac{a}{\delta}} \tag{20}$$

Recall the Poynting vector, $S = \overrightarrow{E} \times \overrightarrow{H}$, and consider the average Poynting vector relationship:

$$S_{avg} \propto |E_y|^2 \tag{21}$$

Since Poynting vector representing the energy, and effect in decibel is normally defined as

$$10log_{10} \frac{P_{original}}{P_{after}} \tag{22}$$

From this understanding, we can naturally approve the following SE formula mentioned in most of books, after refining the formula for this problem:

$$SE = 20log \frac{E_0}{E_2(a)} \tag{23}$$

$$\begin{cases} A = 20log(e^{\frac{a}{\delta}}) \\ R = 20log \frac{Z_1}{4Z_2} \end{cases} \tag{24}$$

3.2.3 B Term

From the Fig. 4, we have known the possibility of multiple reflection issue. Refer to the publication [2], when the electric field encounters a different medium, we have the eq (13) conclusion. With the good conductor assumption, i.e. $Z_2 << Z_1$, we can know that the most of the electric field is reflected at the first boundary. Moreover, after the absorption, for the electric field, the B term is negligible.

However, beyond our assumption of electric field consideration, if we use a magnetic field to evaluate the shielding effectiveness, since the transmission coefficient formula is

$$\frac{2Z_1}{Z_1 + Z_2} \tag{25}$$

We can not simply omit the B term in some specific scenarios, because the first boundary approximately doubles the magnetic field.

4 Simple Calculations and Further Simulations

In Section 3, we have obtained the SE formula (24) with a good conductor assumption. In section 4, we will use the formula (24) to evaluate the SE in an enclosed elevator scenario. Limited to the good conductor assumption, we will utilize a web-based simulation tool [5] for both the elevator scenario and the library room with glass windows scenario (the conductor assumption is not met). The benefit of simulating both scenarios is that we can test whether our theoretical formula (24) is feasible in the 1st scenario.

4.1 Theoretical Calculations for the Elevator Scenario (Good Conductor)

Firstly, we list the necessary equations,

$$\begin{cases} A = 20log(e^{\frac{a}{\delta}}) \\ R = 20log\frac{Z_1}{4Z_2} \end{cases}$$

$$\delta = \sqrt{1/\pi f \mu \sigma}$$

$$Z_2 = \sqrt{\frac{j2\pi f \mu}{\sigma}} \angle 45^\circ$$

With air impedance $Z_1 = 377\Omega$, set the thickness of elevator metal $a \approx 1mm$ ([9]), the material as stainless steel ($\mu = 1000, \sigma = 1.45e6$) ([5]), $f = 1.8kMHz$ supposed 4G network. The results are:

$$\begin{cases} A = 27881.39dB \\ R = 29.57dB \\ A + R = 27910.96dB \end{cases}$$

4.2 Simulations

4.2.1 The Elevator Simulation

Set the parameters of simulator as the section 4.1, we have the following result consistent with the result we get from our theoretical calculations.

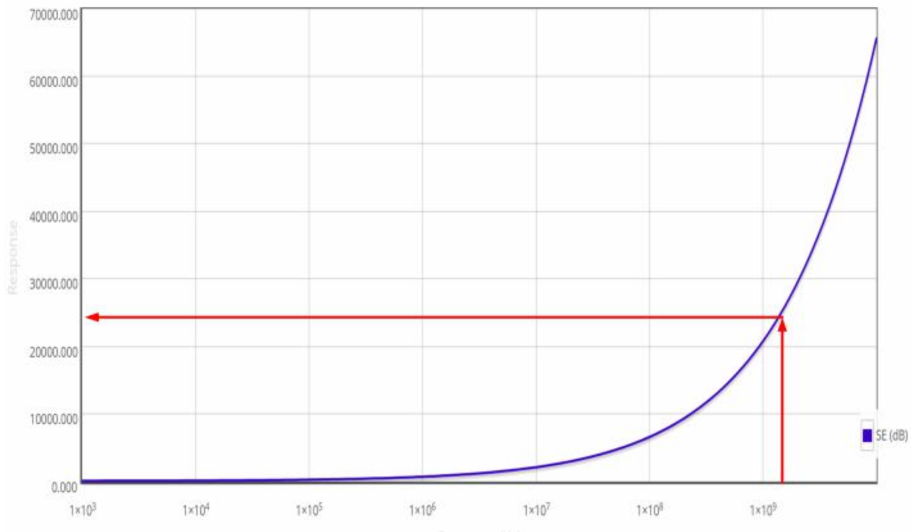

Fig. 6: Elevator

4.2.2 The Library Simulation

By the following simulator settings,

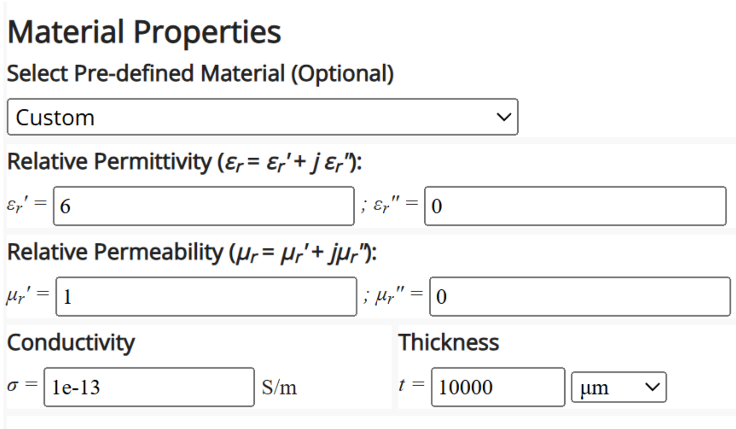

Fig. 7: Setting of Soda-lime glass

We have the SE(f) graph:

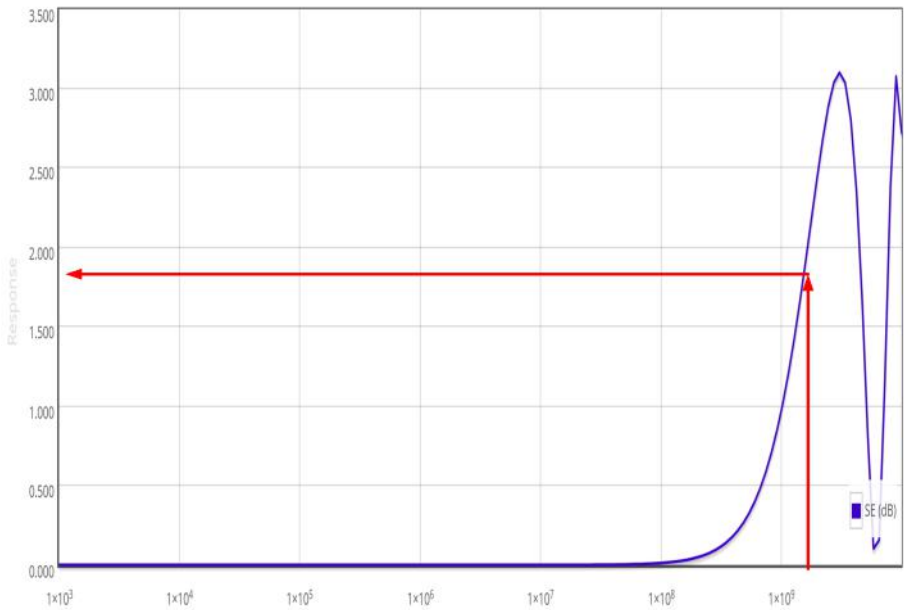

Fig. 8: Soda-lime glass

We can see that the SE value is just $2dB$, which means the glass almost has no effect on the signal. Although the thickness for simulations is $10mm$, ten times larger than the elevator case, this is a counter-intuitive point for people who do not know electromagnetic knowledge.

5 Verification with Real Data and Considerations Based on Reality

In this section, we conduct the real data collection and analyze the gaps between the theoretical calculations and the reality. Firstly, we explain the validation of SE measurements through $RSRP$.

5.1 Conversion from $RSRP$ to SE

The $RSRP$ (Reference Signal Received Power) and Reference Signal Received Quality ($RSRQ$) are key measures of signal level and quality for modern LTE(4G) networks [11]. Simply speaking, from the name and the unit of $RSRP$, "power" and "dBm", we can know there is somehow connection between it and the SE. Technically, if a power P in milliwatts, we have the following definition:

$$RSRP[dBm] = 10log_{10}(P/1mW) \tag{25}$$

With equation (22),

$$RSRP_{original} - RSRP_{after} = 10[(logP_{original} - log(1mW)) - (logP_{after} - log(1mW))]$$

From the above right-hand side, after simplification, we have

$$RSRP_{original} - RSRP_{after} = SE \tag{24}$$

5.2 Elevator Scenario Verification

In the collection process, we utilize the engineering mode of smartphones equipped with IOS. Through the phone function, type the '*3001#12345#*', we can see the signal situation of the smartphone location.

Fig. 9: Engineering Mode

For the *RSRP* just outside the elevator, we collect two data points:

Fig. 10: (a) Outside 1 (b) 2

For the *RSRP* inside the elevator, we collect three data points:

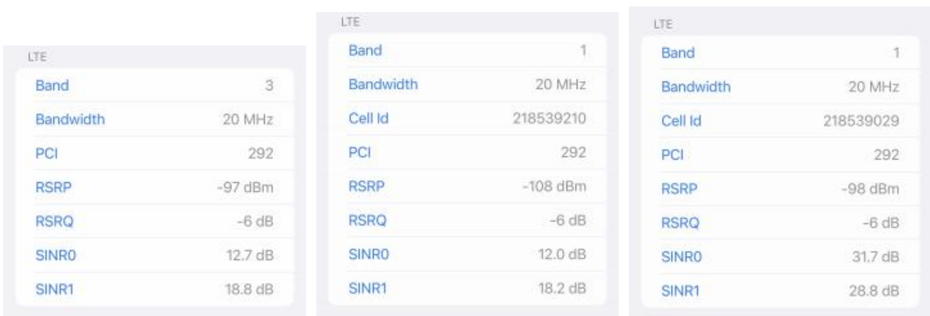

Fig. 11: (a) Inside 1 (b) 2 (c) 3

Taking algebraic mean as the sampled value, we have

$$\begin{cases} RSRP_{outside} = -62dBm \\ RSRP_{inside} = -101dBm \end{cases} \tag{24}$$

5.2.1 The Gap between the Theory and Reality

We have the $SE = 39dB$ which is much smaller than the theoretical calculations. After searching the Internet, we found out that the shielding effect was greatly weakened due to the gaps in the elevator door and other reasons.

5.3 Library Room Scenario Verification

In the room, since the space dimension is much larger than the elevator, we need to collect more data points. However, limited by the smartphone's accuracy, we just collect twenty data points.

The following two figures are data collection process in a library room:

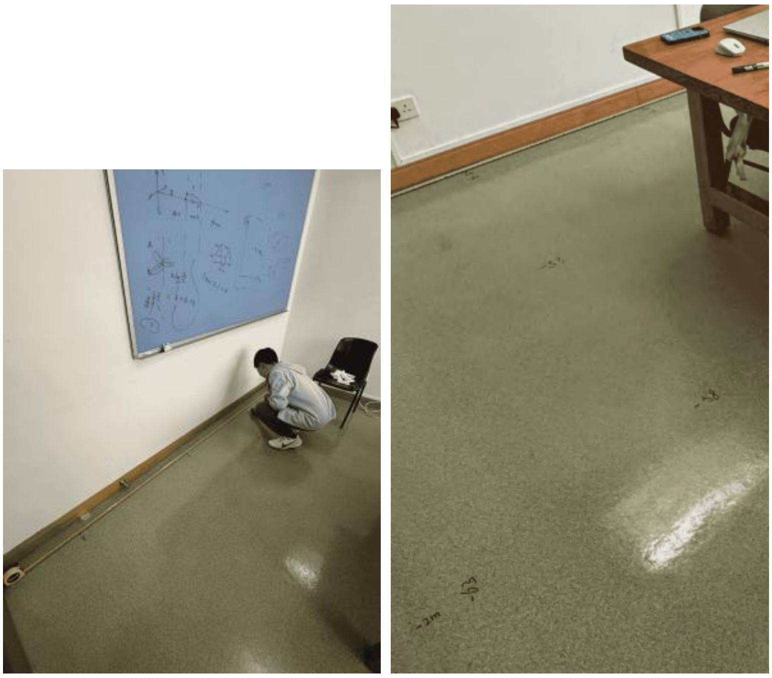

Fig. 12: (a) Coordinate System Establishment (b) *RSRP* Results

The data set is shown in the following two forms:

	0	1	2	3	4
0	-66	-60	-60	-53	-50
1	-61	-53	-50	-58	-56
2	-63	-58	-55	-47	-53
3	-53	-62	-59	-62	-51

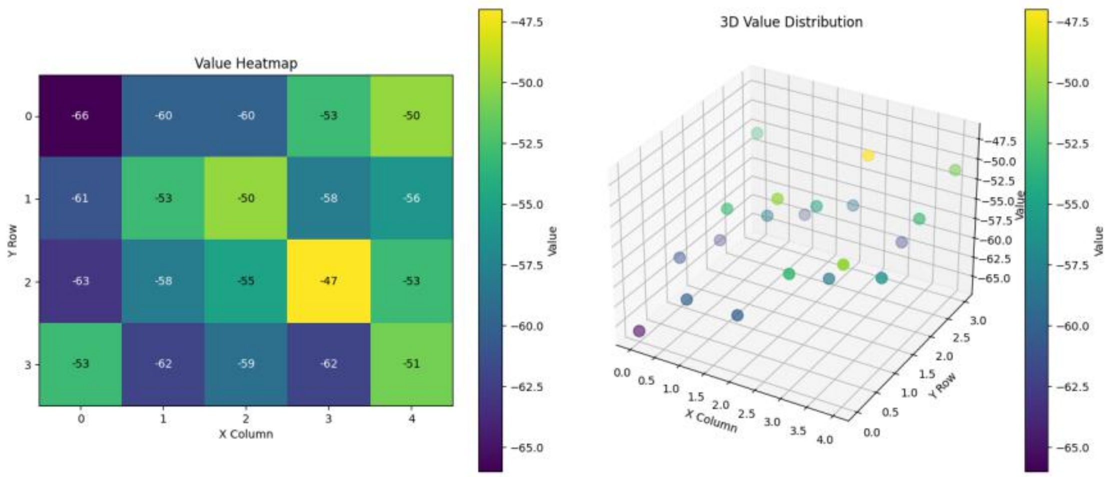

Fig. 13: Inside the Room *RSRP*

For the $RSRP$ outside the room, we take the (5.2).

5.3.1 The Gap between the Theory and Reality

If we directly take the algebraic average as the value inside the room, saying, $-56.5dBm$. Therefore, we can get a negative SE, which means the glass material even enhances the signal, which is obviously unreasonable.

At this point, we search on the Internet again, and know that the multi-path effect always takes place inside a normal room with several reflective surfaces. This phenomenon is very similar to wave interference in affecting the strength of the signal. Since "This superposition may be additive (constructively), destructive (destructively), or both"[7], if we want to strictly analyze the signal distribution, we have to consider it. However, this is too difficult for us to analyze, and to some extent, deviates from the main line of this report to analyze the shielding effect of materials.

In contrast, we can loosely verify the shielding effect of the glass of the library room. Just take the smallest $RSRP$ as the $RSRP_{inside}$, namely, $-66dBm$. As a result, the $SE = 4dB$ which is consistent with the simulation result mentioned in section 4.2.2.

6 Multipath Effect Consideration and Justification for Negligibility

In typical wireless environments, multipath effects arise when signals reflect off surfaces like walls and furniture, creating multiple propagation paths that interfere with the direct signal. These effects typically cause three observable phenomena: signal fading from phase cancellation (described by the Rayleigh fading model), inter-symbol interference due to increased delay spread (τ_{rms}), and reduced SINR (Signal-to-Interference-plus-Noise Ratio).

Our measurements, however, revealed surprisingly high SINR values of 45 dB in elevators and 64 dB in rooms - significantly exceeding the 20 dB threshold typical of multipath-dominated scenarios [8]. This strong signal quality can be quantitatively explained through two key metrics. First, the Rician K-factor:

$$K = \frac{|h_{LOS}|^2}{\sum |h_{NLOS}|^2} \tag{24}$$

where h_{LOS} represents the direct path gain and h_{NLOS} accounts for reflected paths. Our SINR measurements above 45 dB indicate $K \gg 1$, demonstrating overwhelming dominance of the line-of-sight path [1]. Second, the coherence bandwidth:

$$B_c \approx \frac{1}{5\tau_{rms}} \tag{24}$$

With measured delay spreads below 50 ns, we obtain $B_c > 4$ MHz, which comfortably exceeds our 1.4 MHz LTE signal bandwidth [4]. These results collectively show that multipath effects are negligible in our study due to three factors: the strong direct path dominance (high K-factor), minimal delay spread, and absence of deep fades in time-domain measurements.

7 Conclusion

The primary objective of this report was to investigate the fundamental factors influencing electromagnetic shielding, motivated by the common observation of signal intensity fluctuations in everyday environments. Our key findings indicate that, when the signal frequency is fixed, the most significant factors affecting shielding effectiveness are the electric and magnetic properties of the materials encountered. Additionally, our study reveals that signal intensity is influenced not only by material-based shielding, but also by structural features (e.g., small gaps), wave phenomena such as multipath propagation, and other environmental complexities.

These insights enhance our understanding of why signal fluctuations occur and help bridge the gap between fundamental communication theories and electromagnetic physics by tracing the reasoning from Maxwell's equations. However, the report has several limitations. It relies on the plane wave assumption, omits some complex wave behaviors, and involves data collection methods that lack systematic rigor. Future work should incorporate more realistic wave models and adopt more precise and structured data collection to improve the reliability of conclusions.

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